DEPARTMENT OF HEALTH AND HUMAN SERVICES

Food and Drug Administration
Silver Spring MD 20993

IND 113695

MEETING MINUTES

CytRx Corporation
Attention: Molly Salyers
Regulatory Affairs Specialist III
11726 San Vicente Boulevard, Suite 650
Los Angeles, CA 90049

Dear Ms. Salyers:

Please refer to your Investigational New Drug Application (IND) submitted under section 505(i) of the Federal Food, Drug, and Cosmetic Act for aldoxorubicin.

We also refer to the meeting between representatives of your firm and the FDA on Wednesday, March 22, 2017. The purpose of the meeting was

- To discuss the proposal based on aldoxorubicin's nonclinical and clinical safety and doxorubicin efficacy data for demonstration of efficacy and safety in support of a 505(b)(2) application for the treatment of metastatic soft tissue sarcoma; and
- To review the nonclinical and clinical information proposed for submission in the NDA and gain agreement with the Agency that the documentation is adequate to support the submission of the NDA.

A copy of the official minutes of the meeting is enclosed for your information. Please notify us of any significant differences in understanding regarding the meeting outcomes.

If you have any questions, call me at (301) 796-9022.

Sincerely,

{See appended electronic signature page}

Anuja Patel, M.P.H
Senior Project Manager
Division of Oncology Products 2
Office of Hematology and Oncology Products
Center for Drug Evaluation and Research

Enclosure:
Meeting Minutes

FOOD AND DRUG ADMINISTRATION
CENTER FOR DRUG EVALUATION AND RESEARCH

MEMORANDUM OF MEETING MINUTES

Meeting Type: Type C
Meeting Category: Advice, 505(b)(2)

Meeting Date and Time: Wednesday, March 22, 2017
12:30 PM to 1:30 PM, EST
Meeting Location: 10903 New Hampshire Avenue
White Oak Building 22, Conference Room: 1415
Silver Spring, Maryland 20903

Application Number: IND 113695
Product Name: aldoxorubicin
Proposed Indication: Treatment of metastatic soft tissue sarcoma
Sponsor/Applicant Name: CytRx Corporation

Meeting Chair: Dr. Marc Theoret
Meeting Recorder: Anuja Patel

FDA ATTENDEES

Center for Drug Evaluation and Research (CDER), Office of Hematology and Oncology Products (OHOP), Division of Oncology Products 2 (DOP 2)
Patricia Keegan, Director, DOP 2, OHOP
Marc Theoret, Clinical Team Leader, DOP 2, OHOP
Leslie Doros, Medical Reviewer, DOP 2, OHOP
Anuja Patel, Senior Regulatory Health Project Manager, DOP 2, OHOP
Jessica Boehmer, Acting Regulatory Scientist, OHOP

Office of Hematology and Oncology Products (OHOP), Division of Hematology, Oncology, Toxicology (DHOT)
Whitney Helms, Nonclinical Team Leader, DOP 2, DHOT, OHOP
Alex Putman, Nonclinical Reviewer, DOP 2, DHOT, OHOP

Office of Translational Science (OTS), Office of Clinical Pharmacology (OCP), Division of Clinical Pharmacology V (DCP V)
Brian Booth, Deputy Director, DCP V, OCP, OTS, CDER
Hong Zhao, Clinical Pharmacology Team Leader, DCP V, OCP, OTS, CDER
Saeho Chong, Clinical Pharmacology Reviewer, DCP V, OCP, OTS, CDER

Office of Translational Science (OTS), Office of Biostatistics, Biometrics V
Lisa Rodriguez, Biostatistics Team Leader, OB V, OTS, CDER
Vivian Weishi, Biostatistics Reviewer, OB V, OTS, CDER

Office of Product Quality (OPQ), Office of New Drug Products (ONDP), Division of New Drug Products 1 (DNP 1), CDER
Joyce Crich, Quality Assurance Lead, DNP 1, ONDP, OPQ, CDER

Office of Program and Regulatory Operations (OPRO)
Steven Kinsley, Regulatory Business Process Manager

SPONSOR ATTENDEES

CytRx Corporation:
Daniel Levitt, M.D., Ph.D., Executive Vice President, Chief Medical Officer
Scott Wieland, Ph.D., Senior Vice President, Drug Development
Mary (Molly) Salyers, Regulatory Affairs Specialist III
Shanta Chawla, M.D., Vice President of Clinical Development

Hurley Consulting Associates Ltd.:
Margaret E. Hurley, M.D., President and CEO
John C. Talian, Ph.D., Vice President, Regulatory Affairs
Heather Knight-Trent, Pharm.D., Executive Director, Regulatory Affairs
Charles G. Garlisi, Ph.D., Vice President, Scientific Affairs
Joseph P. Tizzano, Ph.D., Director, Toxicology

BACKGROUND

Meeting Purpose

In the February 21, 2017, cover letter to the meeting briefing package, CytRx states that the purpose of this meeting is:

- "To discuss the proposal based on aldoxorubicin's nonclinical and clinical safety and doxorubicin efficacy data for demonstration of efficacy and safety in support of a 505(b)(2) application for the treatment of metastatic soft tissue sarcoma."
- "To review the nonclinical and clinical information proposed for submission in the NDA and gain agreement with the Agency that the documentation is adequate to support the submission of the NDA."

As stated in the meeting briefing package, the proposed indication is for "the treatment of metastatic soft tissue sarcoma."

The proposed dosage regimen is 350 mg/m^2 of aldoxorubicin (doxorubicin equivalent dose of 260 mg/m^2) administered intravenously over 30 minutes once every 3 weeks (21 days).

Regulatory

The key regulatory interactions are summarized below:

The development of aldoxorubicin was conducted under INDs 75478, 113695, and 114651. IND 113695 was submitted on October 20, 2011, for the development program in soft tissue sarcoma (STS).

- On November 18, 2011, IND 113695 was allowed to proceed for the conduct of Protocol INNO-206-P2-STS-01, titled, "A Multicenter, Randomized, Open-Label Phase 2b Study to Investigate the Preliminary Efficacy and Safety of INNO-206 (Doxorubicin-EMCH) Compared to Doxorubicin in Subjects with Metastatic, Locally Advanced, or Unresectable Soft Tissue Sarcoma."

- On October 29, 2012, FDA held an end-of-phase 2 (EOP2) meeting to discuss the adequacy of a single clinical trial, ALDOXORUBICIN-P3-STS-01 titled "A Multicenter, Randomized, Open-Label Phase 3 Study to Investigate the Efficacy and Safety of Aldoxorubicin Compared to Dacarbazine in Subjects with Metastatic, Locally Advanced, or Unresectable Soft Tissue Sarcomas Who Either Relapsed or Were Refractory to Prior Non-Adjuvant Chemotherapy" to support an NDA for aldoxorubicin for the treatment of patients receiving second or third-line treatment for soft tissue sarcoma (STS). FDA agreed that the proposed trial could support an NDA or efficacy supplement provided that the trial demonstrated a robust, highly statistically significant and clinically important improvement in overall survival. FDA stated that they could not agree that a treatment effect on PFS could support a marketing application since there was insufficient information on the effect size to be detected and timing of the PFS analysis. FDA also noted that a safety database of 300 to 600 patients is adequate to support a marketing application when the product will be administered for 6 months at dosage levels intended for clinical use.

- On January 23, 2013, CytRx submitted an SPA request for the ALDOXORUBICIN-P3-STS-01 study. On February 26, 2013, FDA held a teleconference with CytRx to discuss deficiencies in the proposed protocol. On March 4, 2013, FDA issued an SPA Non-Agreement letter, which noted, among other things, that the targeted magnitude of improvement (difference in median PFS of 2.1 months between arms) was unlikely to be clinically meaningful.

- On April 20, 2013, FDA issued an SPA Agreement letter for the revised Protocol ALDOXORUBICIN-P3-STS-01, received on April 4, 2013, and the revised SAP, received April 4, 2013. In the April 20, 2013, letter, FDA stated "that a robust, highly statistically significant improvement in PFS that is of sufficient magnitude to be considered clinically meaningful such that the risk benefit profile for the drug is favorable, may support a future regulatory application. As we stated during the February 26, 2013, teleconference, the magnitude of improvement in PFS that would be considered clinically meaningful is dependent upon the magnitude of the PFS treatment effect observed in the clinical trial and the risk profile of aldoxorubicin. This would need to include results of the overall survival analysis to ensure no decrement in overall survival for aldoxorubicin-treated patients. We also stated that the exact magnitude of PFS improvement that would be considered significant for clinical benefit is a review issue and a final decision can only be made based on the review of the submitted data."

- On April 29, 2014, CytRx submitted a request for Breakthrough Therapy Designation (BTD) for the treatment of advanced metastatic STS based upon efficacy results from trial INNO-206-P2-STS-01. On June 16, 2014, FDA issued a BTD denial letter, stating that the magnitude of improvement in PFS over doxorubicin was modest and not consistent across histologic subtypes of STS.

- On October 9, 2015, CytRx submitted a second request for BTD for the treatment of STS with an improved cardiac safety profile as compared to doxorubicin; this request was denied on December 9, 2015, BTD denial letter. The basis for this denial was summarized as follows: (1) the trial submitted to provide evidence of cardiac safety compared to doxorubicin was inadequate in design to provide an assessment of the reduction in cardiotoxicity as the trial lacked a pre-specified objective and analysis plan to test that hypothesis, (2) while there was a modest difference between trial arms in the echocardiogram results post-treatment, favoring the aldoxorubicin arm, the clinical relevance of this finding was unclear as no patient in either arm experienced clinically relevant cardiac adverse events (AEs) that required modification of doxorubicin or aldoxorubicin, (3) cardiac AEs considered at least possibly related to study drug were only reported in the aldoxorubicin arm, and (4) the lack of reports of congestive heart failure in trials of aldoxorubicin-treated patients was interpreted in the context of short duration of follow-up for the majority of patients.

- On December 13, 2016, CytRx submitted a pre-NDA, type B meeting request. The meeting was granted as a Type B meeting on December 29, 2016, and the meeting background was received on February 21, 2017. Based on the purpose of the meeting, the information and questions for FDA included in the February 21, 2017, meeting package, and the absence of the top-line results of human studies supporting a scientific bridge between aldoxorubicin and doxorubicin HCl or studies which establish the effectiveness of aldoxorubicin, FDA changed the meeting type to a Type C/Advice meeting.

FDA sent Preliminary Meeting Responses to CytRx on March 21, 2017.

Proposed Content of a Planned NDA

CytRx proposes to relay on FDA's prior findings of safety and effectiveness for doxorubicin and data obtained with aldoxorubicin, as follows, in support of an NDA submitted under the provisions of 505(b)(2):

- Chemistry, Manufacturing, and Controls
- Aldoxorubicin

- Nonclinical
- Aldoxorubicin: pharmacology, safety pharmacology, pharmacokinetics, and toxicology
- Doxorubicin: carcinogenicity and reproductive studies
- Doxil (doxorubicin hydrochloride liposome injection) USPI: carcinogenicity and reproductive studies (pg 53/477)
- Clinical
- Doxorubicin: drug-drug interactions including CYP enzymes and transporters;
  Doxorubicin: efficacy data in metastatic soft tissue sarcoma; and
- Aldoxorubicin: pharmacokinetic data from Phase 1 studies and safety data from Phase 1, 2, and 3 studies.

Chemistry, Manufacturing, and Control

The drug substance aldoxorubicin has the chemical name (E)-N'-(1-((2S,4S)-4-(4-amino-5-hydroxy-6-methyltetrahydro-2H-pyran-2-yloxy-2,5,12-trihydroxy-7-methoxy-6,11-dioxo-1,2,3,4,6,11-hexahydrotetracen-2-yl)-2-hydroxyethylidene)-6-(2,5-dioxo-2H-pyrrol-1(5H)-yl)hexanehydrazide hydrochloride. Aldoxorubicin is the (6-maleimidocaproyl) hydrazone of doxorubicin, a prodrug of doxorubicin. The molecular formula is $C_{37}H_{42}N_4O_{13}\cdot HCl$, and the molecular weight is 787.22 g/mol. The drug substance is a red powder that is soluble in aqueous buffers pH 5.0 to 7.5. The drug product is 200 mg of a lyophilized powder supplied in clear glass vials with Teflon stoppers. The drug product vials are stored at -20 °C and protected from light. The drug product is reconstituted with 20 mL of 50% ethanol-water (10 mg/mL) and further diluted into Lactated Ringers' Solution for IV administration.

Nonclinical

CytRx describes pharmacology data demonstrating the anti-cancer activity of aldoxorubicin in various mouse models of cancer. The non-clinical program for aldoxorubicin consists of repeated-dose toxicity studies in rats and dogs, a cardiovascular and respiratory safety pharmacology study in dogs, and a full genetic toxicology battery. Reproductive toxicity studies will not be conducted. CytRx will also rely on the safety established in the registration of doxorubicin HCL.

Clinical Pharmacology

Two pharmacokinetic (PK) studies of aldoxorubicin have been conducted in patients with advanced solid tumors. The study conducted at Tumor Biology Institute in Germany was a dose escalation study in 41 subjects who received aldoxorubicin as a 30-minute infusion on Day 1 of a 21-day cycle at dose levels of 27 to 460 mg/m^2. The total doxorubicin concentrations were determined in plasma samples collected up to 72 hours and in urine samples collected up to 24 hours after the first dose of aldoxorubicin. CytRx states that in general, Cmax and AUC increased proportionally with the dose, although the dose groups receiving 200, 270, and 460 mg/m^2 aldoxorubicin did not fit to the overall trend. There was no correlation between clearance and the dose of aldoxorubicin administered. The fraction of doxorubicin excreted in the urine was 1.4% to 7.7% of the total amount of aldoxorubicin administered. The maximum tolerated dose (MTD) of aldoxorubicin was determined to be 340 mg/m^2.

The ALDOXORUBICIN-P1-PK-01 study conducted in 18 subjects at dose of 230 mg/m^2 and 350 mg/m^2 (165 and 260 mg/m^2 doxorubicin equivalents, respectively) was a PK and safety study of a new formulation of aldoxorubicin. Subjects were administered aldoxorubicin by intravenous (IV) infusion on Day 1 every 21 days for up to 8 consecutive cycles. The complete PK profile was determined during Cycle 1 and a limited PK profile was evaluated during Cycle 3. Doxorubicin and albumin-bound aldoxorubicin concentrations were analyzed separately. Per CytRX, albumin-bound doxorubicin (i.e., aldoxorubicin) comprised over 99% of total doxorubicin in the circulation over a 72 hour period and the major doxorubicin metabolite, doxorubicinol, was <0.0007% of total doxorubicin. CytRx states that aldoxorubicin was bound to circulating albumin in less than 5 minutes and demonstrated a 20 hour half-life with clearance that was 180-fold lower than that of doxorubicin, and a steady state volume of distribution 140-fold lower. CytRx states that these results are distinctly different than the PK of doxorubicin when doxorubicin is administered. Over a 72 hour period, only 1% to 2% of total doxorubicin and 0.001% to 0.003% of doxorubicinol could be detected in the urine.

Based on the meeting package, no other clinical pharmacology studies of aldoxorubicin have been conducted.

Clinical

<u>Overview of all Clinical Studies with Aldoxorubicin</u>

According to the meeting package, there are 12 clinical trials evaluating the safety and efficacy of aldoxorubicin. CytRx states that the NDA will rely on safety data obtained in all of the trials listed below for characterization of the safety of aldoxorubicin. The proposed NDA will rely on clinical studies with aldoxorubicin conducted in patients with metastatic soft tissue sarcoma (STS) and on FDA's finding of effectiveness for doxorubicin HCL to demonstrate the efficacy of aldoxorubicin in support of this 505(b)(2) application. Table 1 summarizes ongoing and completed studies of aldoxorubicin.

Table 1. Summary of Ongoing and Completed Studies with Aldoxorubicin

Study ID	Indication	Study design	Pts	Results	Safety Data
Tumor Biology Institute in Germany	Advanced solid tumor	Phase 1 dose-escalation clinical study to determine the MTD	41	The MTD and RP2D was 350 mg/m^2.	No clinical signs CHF. No decrease in LVEF. DLTs at 350 mg/m^2 were neutropenic fever and mucositis.
INNO-206-P1-MTD-01	Advanced solid tumor who received 1 or more prior therapies	Phase 1b/2 open-label establish MTD of new formulation every 21 days for up to 8 cycles	25	MTD was determined to be 350 mg/m^2. Of the 13 evaluable patients with STS, 5 objective PR have been documented as well as 7 patients with SD (range 1-10+ months).	
ALDOXORUBICIN-P1-MTD-02	Solid tumors	Phase 1b/2 open-label combination of aldoxorubicin and doxorubicin preliminary safety and MTD. Patients received 35 mg/m^2 doxorubicin with escalating doses of aldoxorubicin	15	Tumor responses were observed in subjects in metastatic breast cancer and STS. The MTD was 320 mg/m^2 aldoxorubicin + 35 mg/m^2 doxorubicin.	DLTs included grade 4 mucositis and grade 4 fatigue.
ALDOXORUBICIN-P1-MTD-03	Metastatic solid tumor	Phase 1b preliminary safety and activity of aldoxorubicin plus gemcitabine	30	Completed	Summary not provided
ALDOXORUBICIN-P1/2-STS-03	Metastatic or locally advanced STS	Phase 1/2 single-center, open-label preliminary safety and activity of aldoxorubicin plus ifosfamide	30	Ongoing	Summary not provided
ALDOXORUBICIN-P1-PK-01	Advanced solid tumor	Phase 1 open-label to evaluate the PK and safety of aldoxorubicin. Aldoxorubicin was administered at doses of 230 mg/m^2 and 350 mg/m^2 on D1 every 21 days for up to 8 consecutive cycles.	18	A minor response (25% decrease, 230 mg/m^2) was observed in 1 patient with SCLC who had received 3 prior treatments. This patient received 21 cycles (230 mg/m^2/cycle, 4,830 mg/m^2 cumulative dose of aldoxorubicin, 3,575 mg/m^2 cumulative dose of doxorubicin equivalents) with no evidence of cardiotoxicity. Three other patients received more than 10 cycles of drug. The PK parameters of aldoxorubicin of administered doses up to MTD of 350 mg/m^2 (doxorubicin equivalent dose of 260 mg/m^2) in advanced cancer patients suggest that the drug is stable in circulation and does not accumulate to a substantial degree in body compartments outside of the bloodstream.	One SAE of febrile neutropenia, was reported and 2 deaths due to tumor progression occurred. Grades 3 and 4 AEs included neutropenia, anemia and thrombocytopenia. A minor response (25% decrease, 230 mg/m2) was observed in 1 patient with SCLC who had received 3 prior treatments. This patient received 21 cycles (4,830 mg/m2 cumulative dose) of aldoxorubicin, with no evidence of cardiotoxicity. Three other patients received more than 10 cycles of drug.
INNO-206-P2-STS-01	Metastatic locally advanced, unresectable STS	Phase 2b, randomized, open-label, prospective, multicenter to evaluate the efficacy and safety of aldoxorubicin 350 mg/m^2 compared to doxorubicin 75 mg/m^2 on D1 of 21 day cycle. Maximum of 6 cycles. Primary Endpoint: PFS Secondary endpoints: OS, PFS at 4	126	Median PFS was 5.6 months in the aldoxorubicin arm vs. 2.7 months in the doxorubicin arm (stratified log-rank p value=0.02). There was no difference in median OS (15.8 vs. 14.3 months; stratified log-rank p-value=0.21) and ORR was 25% in the aldoxorubicin arm and 0% in the	There was an increase in the incidence of AEs in the aldoxorubicin group compared to the doxorubicin group, but no new AEs emerged.

Study ID	Indication	Study design	Pts	Results	Safety Data
		and 6 months, ORR, and the frequency of treatment-related toxicity		doxorubicin arm.	
INNO 206 P2-PDA-01	Advanced Pancreatic ductal carcinoma with progression after treatment with 2 prior regimens, the first containing gemcitabine and the second containing fluoropyrimidine.	Phase 2 multicenter, open-Label, pilot study to evaluate the preliminary efficacy and safety of INNO-206 administered at 350 mg/m^2 IV every 21 days for up to 8 consecutive cycles.	14	Completed	Summary not provided
ALDOXORUBICIN-P2-KS-01	HIV patients with Kaposi's sarcoma	Phase 2 pilot study was conducted to investigate efficacy and intratumoral kinetics (primary objectives) and safety (secondary objective) of aldoxorubicin	28	Completed	Summary not provided
ALDOXORUBICIN-P2-GBM-01	Unresectable GBM who progressed after treatment with surgery, radiation, and temozolomide	Phase 2 open-label pilot study efficacy of aldoxorubicin.	28	Completed	Summary not provided.
ALDOXORUBICIN-P2-SCLC 01	Metastatic SCLC who either relapsed or were refractory to prior chemotherapy	Phase 2b multicenter, randomized, open-label investigating the efficacy and safety of aldoxorubicin compared to topotecan subjects	132	Ongoing	Summary not provided
ALDOXORUBICIN-P3-STS-01	Metastatic, locally advanced, or unresectable STS who have relapsed or were refractory to prior non-adjuvant chemotherapy	Phase 3 randomized, multi-center, open-label trial comparing aldoxorubicin to investigator's choice. Aldoxorubicin was administered 350mg/m2 IV on D1 of each 21-day cycle. Primary endpoint was PFS, Secondary endpoints: OS,ORR, disease control rate, PFS at 4 and 6 months, and the frequency of treatment-related toxicities.	433	This study did not demonstrate a statistically significant difference in overall PFS for aldoxorubicin compared to Investigator's Choice (HR: 0.82, p value = 0.124) or a statistically significant OS difference for aldoxorubicin versus Investigator's Choice (HR: 1.04, p value = 0.809) at the interim analysis when the primary analysis of PFS was completed.	Adverse events reports for subjects in the aldoxorubicin arm were chiefly stomatitis and those associated with bone marrow suppression such as anemia, neutropenia, febrile neutropenia and thrombocytopenia. The effect of aldoxorubicin on LVEF appears similar to that of doxorubicin despite greater exposure based on doxorubicin equivalents. Dose-limiting cardiovascular adverse events were not observed with aldoxorubicin at a median cumulative doxorubicin equivalent dose of 1,040 mg/m2 (range: 260 to 7,800 mg/m2).

Abbreviations: Pts: patients, MTD: maximum tolerated dose, RP2D: recommended phase 2 dose, CHF: congestive heart failure, LVEF: left ventricular ejection fraction, DLT: dose-limiting toxicity, STS: soft tissue sarcoma, PR: partial response, SD: stable disease, PK: pharmacokinetics, SAE: serious adverse event, AE: adverse event, SCLC: small cell lung cancer,, D: day, PFS: progression-free survival, OS: overall survival, ORR: overall response rate, HR: hazard ratio.

Source: Data from the February 21, 2017 meeting package

The two clinical trials of aldoxorubicin most relevant to the proposed indication are Protocols INNO-206-P2-STS-01 and ALDOXORUBICIN-P3-STS-01. Summary information for each trial based on the meeting package is provided below.

Protocol INNO-206-P2-STS-01

Design

Protocol INNO-206-P2-STS-01 was a randomized, open-label, multicenter, active-controlled trial in patients with metastatic or locally advanced, unresectable STS that evaluated the efficacy and safety of aldoxorubicin compared with doxorubicin. Prior exposure to doxorubicin or liposomal doxorubicin less than 375 mg/m^2 were eligible to enroll. Patients were randomly assigned in a 2:1 ratio to either aldoxorubicin 350 mg/m^2 or doxorubicin 75 mg/m^2 arms, which were administered on Day 1 of each 21-day cycle. Treatment was given for a maximum of six cycles.

The primary endpoint was progression-free survival (PFS) as assessed by blinded independent central review (BICR) per RECIST v1.1 in the modified intent-to-treat population, defined as all patients who received at least one dose of aldoxorubicin or doxorubicin and had at least one post-baseline tumor assessment on study. Secondary endpoints were overall survival (OS), PFS at 4 and 6 months, overall response rate (ORR), and the frequency of treatment-related toxicity. The PFS and tumor response endpoints were assessed both by local investigators and by independent blinded central review. Toxicities were graded by NCI CTCAE v4.0.

Assuming that the median PFS was 4.4 months in the doxorubicin arm and of 8.5 months in the aldoxorubicin arm, a total of 89 events are needed to detect a hazard ratio of 0.52 with 83% power at a two-sided alpha level of 5%. Assuming an 18 month accrual period and 12-month follow-up, a total of 105 were required

Results

The trial enrolled 126 patients with 83 patients randomized to the aldoxorubicin arm and 43 patients randomized to the doxorubicin arm. The median numbers of cycles were six (range one to eight cycles) for the aldoxorubicin group and four (range one to six cycles) for the doxorubicin group. The primary efficacy analysis of IRC-assessed PFS was conducted in 118 patients in the modified ITT population, which excluded 3 (4%) patients from the aldoxorubicin arm and 5 (12%) patients in the doxorubicin arm. The analysis of OS excluded 7% (3/43) of patients randomized to doxorubicin. A summary of the efficacy results are presented in the table below.

Table 2. Efficacy Results for Trial INNO-206-P2-STS-01

	Central Assessment		Investigator Assessment	
	Aldoxorubicin N=80	Doxorubicin N=38	Aldoxorubicin N=83	Doxorubicin N=40
PFS				
Median, months	5.6	2.7	8.3	4.6
HR (95% CI)[a]	0.6 (0.38, 0.95)		0.44 (0.27, 0.71)	
P-value[b]	0.0228		0.0006	
Overall Survival[c]				
Median, months			15.8	14.3
HR (95%, CI)[d]			0.73 (0.44, 1.20)	
P-value[b]			0.215	
Objective Response Rate				
% (95% CI)	25 (16.0, 35.9)	0.0 (0.0, 9.3)	23 (14.4, 33.4)	5 (0.6, 16.9)

[a] Hazard ratio and 95% CI are from a Cox regression model stratified by ECOG PS (0-1, 2) and chemotherapy (no prior chemotherapy or prior adjuvant/neoadjuvant chemotherapy).
[b] P-value is from 2-sided stratified log-rank test with ECOG PS (0 to 1, 2) and chemotherapy (no prior chemotherapy or prior adjuvant/neoadjuvant chemotherapy) as stratification factors.
[c] The efficacy subpopulation comprised all subjects who received at least 1 dose of aldoxorubicin or doxorubicin and had at least 1 post-baseline tumor assessment on study.
[d] Two-sided 95% CIs for each treatment group are estimated using the Kaplan-Meier method; 2-sided 95% CIs for the difference between treatment groups are estimated by approximate normal distributions described in the SAP.
Source: Data from the February 21, 2017 meeting package

CytRx states that the safety analysis in this study demonstrated an increase in the incidence of adverse events (AEs) reported in the aldoxorubicin group compared to the doxorubicin group. According to the meeting package, no new adverse events of potential concern were identified with the increased exposure in the aldoxorubicin group.

Table 3. Summary of Treatment Emergent Adverse Events (Safety Population[a]) Trial INNO-206-P2-STS-01

Category of Adverse Event, n (%)	**Aldoxorubicin (n = 83)**	**Doxorubicin (n = 40)**
Subjects With Any TEAE	79 (95.2)	32 (80.0)
Subjects With Any Severe or Worse TEAE[b]	51 (61.4)	18 (45.0)
Subjects With Any TEAE Leading to Study Drug Discontinuation or Interruption	30 (36.1)	7 (17.5)
Subjects With Any Serious TEAE	33 (39.8)	8 (20.0)
Subjects With Any TEAE Resulting in Death	0	1 (2.5)

Adverse events were coded using MedDRA Version 14.1.
MedDRA = Medical Dictionary for Regulatory Activities; TEAE = treatment-emergent adverse event
[a] The safety population comprised all subjects who received any amount of aldoxorubicin or doxorubicin on study.
[b]Adverse events were those with CTCAE Grade ≥3.

Source: Data from the February 21, 2017 meeting package

Trial ALDOXORUBICIN-P3-STS-01

Design

Study ALDOXORUBICIN-P3-STS-01 is a multicenter, open-label, randomized controlled trial comparing aldoxorubicin to Investigator's Choice in patients with metastatic, locally advanced, or unresectable STS who relapsed or were refractory to prior non-adjuvant chemotherapy. The randomization was stratified according to ECOG 0-1 vs 2, number of prior regimens (1 or >1), prior doxorubicin, and histologic classification according to local review (liposarcoma, leiomyosarcoma, synovial sarcoma, and other sarcomas). Eligible patients were randomly assigned in a 1:1 ratio to the following treatment arms:

1. Experimental arm: Aldoxorubicin administered at 350 mg/m^2 (260 mg/m^2 doxorubicin equivalent) IV on Day 1 every 21 days for up to 8 cycles

2. Investigator's Choice arm:

   - Dacarbazine administered 1000 mg/m^2 intravenously (IV) on Day 1 every 21 days for up to 8 consecutive cycles;
   - Pazopanib, 800 mg orally daily until either tumor progression or unacceptable toxicity occurs;
   - Gemcitabine, 900 mg/m^2 IV on Days 1 and 8, plus docetaxel, 100 mg/m^2 IV on Day 8 of a 28 day cycle for up to 8 cycles;
   - Doxorubicin, 75 mg/m^2 IV every 21 days for up to 8 cycles or a maximum cumulative dose of 550 mg/m^2; or
   - Ifosfamide 2 gm/m^2 IV on Days 1 to 4 of a 21 day cycle with mesna for up to 8 cycles.

The investigator site pre-specified their choice of comparator arms. The primary endpoint of PFS was assessed by BICR per RECIST v1.1. The secondary endpoints included OS, ORR, disease control rate, PFS at 4 and 6 months, and the frequency of treatment-related toxicities in this population. Toxicities were graded by NCI CTCAE v4.03.

Assuming a median PFS of 3.5 months for the Investigator's Choice arm and 5.6 months for the aldoxorubicin arm, a total of 191 PFS events was required for 90% power at a 2-sided 5% alpha level to detect this difference. Approximately 400 subjects would be needed to achieve the total of 191 PFS events. For the secondary endpoint of OS, the median OS for the Investigator's Choice arm was estimated to be 10 months. The final analysis of OS will occur when there are 320 deaths. Based on the two-sided log rank test at the α=0.049 level of significance, there would be 85% power to detect a statistically significant difference if the median survival in the aldoxorubicin group is 14 months. The assumed medians corresponded to a hazard ratio of 0.71.

Results

The results of this study did not demonstrate a statistically significant difference in PFS for aldoxorubicin compared to Investigator's Choice for the intent-to-treat population (HR: 0.82, p value = 0.124).

The study is continuing for OS which will be completed when 320 deaths have occurred. The study did not demonstrate a statistically significant OS difference for aldoxorubicin versus Investigator's Choice (HR: 1.04, p value = 0.809) at the interim analysis of OS.

Table 4. Efficacy Results, Trial ALDOXORUBICIN-P3-STS-01

	Aldoxorubicin N=218	**Investigator Choice N= 215**
PFS		
Median, months	4.04	2.96
HR (95% CI)	0.82 (0.64, 1.06)	
P-value	0.1244[a]	
Overall Survival		
Median, months	12.6	12.0
HR (95%, CI)	1.04 (0.76, 1.42)	
P-value	0.809[a]	
Objective Response Rate		
% (95% CI)	7.3 (4.25, 11.65)	4.2 (1.93, 7.80)

[a] P-value is based on 2-sided, unstratified, log-rank test.
Source: Data from the February 21, 2017 meeting package

The preliminary overviews of AEs are summarized in Table 5.

Table 5. Overview of Adverse Events, Trial ALDOXORUBICIN-P3-STS-01

Adverse Event Category	**Aldoxorubicin (N = 213) n (%)**	**Investigator's Choice (N = 207) n (%)**
Number of Subjects With at Least 1 TEAE[a]	209 (98.1)	203 (98.1)
Grade ≥3 Adverse Events	158 (74.2)	133 (64.3)
Adverse Events Leading to Death	7 (3.3)[b]	1 (0.5)
Adverse Events Leading to Study Drug Discontinuation	23 (10.8)	21 (10.1)
Serious Adverse Events	91 (42.7)	68 (32.9)
Adverse Events by Intensity[a]		
Grade 1	15 (7.0)	27 (13.0)
Grade 2	36 (16.9)	43 (20.8)
Grade 3 or 4	151 (70.9)	132 (63.8)
Grade 5	7 (3.3)	1 (0.5)

[a] Source data on file.
[b] Two patients were identified as having adverse events leading to death. However, both patients had disease progression and should not have been categorized as adverse events leading to death.

Source: Data from the February 21, 2017 meeting package

Seven (3.3%) deaths on the aldoxorubicin arm were reported due to AEs of which three deaths (1.4%) were designated as treatment-related (death unknown cause, sepsis, pneumonia). The most common AEs leading to discontinuation of aldoxorubicin were peripheral neuropathy in 12 (6%) patients and neutropenia in two (< 1%) patients in the aldoxorubicin arm. One patient discontinued aldoxorubicin due to decreased LVEF. Grade 3 or 4 treatment emergent adverse events (TEAEs) in >5% of patients treated with aldoxorubicin were neutropenia (24%), anemia (22%), febrile neutropenia (16%), thrombocytopenia (10%), stomatitis (16%), white blood cell count decreased (11%), and hypophosphatemia (5%). Other significant AEs that occurred with aldoxorubicin were neuropathy (15%) and sepsis (7%).

A total of 22 (10%) patients receiving aldoxorubicin reported at least one cardiac AE with no Grade 5 occurrences. One (0.5%) patient experienced two Grade 4 AEs of pulseless electrical activity and cardiac arrest. Both AEs were deemed unrelated to aldoxorubicin; the outcomes were recovered and resolved. A ≥20% decrease in LVEF occurred in eight patients (3.8%) and acute cardiac failure and cardiomyopathy occurred in one patient each.

DISCUSSION

FDA General Comments

FDA understands that CytRx intends to submit a 505(b)(2) application which will rely on FDA's finding of effectiveness for doxorubicin for the proposed indication (the indication in the doxorubicin HCl labeling is "indicated for the treatment of metastatic soft tissue sarcoma"). In the meeting package, CytRx states that it will demonstrate the pharmaceutical equivalence of aldoxorubicin with doxorubicin in accordance with the FDA Guidance for 505(b)(2) applications. In addition, CytRx also states that they intend to rely on statements made in the FDA-approved package insert for Doxil.

The Division recommends that CytRx considering the submission of an application through the 505(b)(2) pathway consult the Agency's regulations at 21 CFR 314.54, and the draft guidance for industry, *Applications Covered by Section 505(b)(2)* (October 1999), available at http://www.fda.gov/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/default.htm. In addition, FDA has explained the background and applicability of section 505(b)(2) in its October 14, 2003, response to a number of citizen petitions that had challenged the Agency's interpretation of this statutory provision (see Docket FDA-2003-P-0274-0015, available at http://www.regulations.gov).

If CytRx intends to submit a 505(b)(2) application that relies for approval on FDA's finding of safety and/or effectiveness for one or more listed drugs, CytRx must establish that such reliance is scientifically appropriate and must submit data necessary to support any aspects of the proposed drug product that represent modifications to the listed drug(s). CytRx should establish a "bridge" (e.g., via comparative bioavailability data) between aldoxorubicin and each listed drug upon which CytRx proposes to rely to demonstrate that such reliance is scientifically justified.

If CytRx intends to rely on literature or other studies for which CytRx has no right of reference but that are necessary for approval, CytRx also must establish that reliance on the studies described in the literature or on the other studies is scientifically appropriate. CytRx should include a copy of such published literature in the 505(b)(2) application and identify any listed drug(s) described in the published literature (e.g. by trade name(s)).

If CytRx intends to rely on the Agency's finding of safety and/or effectiveness for a listed drug(s) or published literature describing a listed drug(s) (which is considered to be reliance on FDA's finding of safety and/or effectiveness for the listed drug(s)), CytRx should identify the listed drug(s) in accordance with the Agency's regulations at 21 CFR 314.54. It should be noted that 21 CFR 314.54 requires identification of the "listed drug for which FDA has made a finding of safety and effectiveness," and thus an applicant may only rely upon a listed drug that was approved in an NDA under section 505(c) of the FD&C Act. The regulatory requirements for a 505(b)(2) application (including, but not limited to, an appropriate patent certification or statement) apply to each listed drug upon which CytRx relies.

If FDA has approved one or more pharmaceutically equivalent products in one or more NDA(s) before the date of submission of the original 505(b)(2) application, CytRx must identify one such pharmaceutically equivalent product as a listed drug (or an additional listed drug) relied upon (see 21 CFR 314.50(i)(1)(i)(C), 314.54, and 314.125(b)(19); see also 21 CFR 314.101(d)(9)). If CytRx identify a listed drug solely to comply with this regulatory requirement, CytRx must provide an appropriate patent certification or statement for any patents that are listed in the Orange Book for the pharmaceutically equivalent product, but CytRx is not required to establish a "bridge" to justify the scientific appropriateness of reliance on the pharmaceutically equivalent product if it is scientifically unnecessary to support approval.

If CytRx propose to rely on FDA's finding of safety and/or effectiveness for a listed drug that has been discontinued from marketing, the acceptability of this approach will be contingent on FDA's consideration of whether the drug was discontinued for reasons of safety or effectiveness.

FDA encourages CytRx to identify each section of the proposed 505(b)(2) application that is supported by reliance on FDA's finding of safety and/or effectiveness for a listed drug(s) or on published literature (see table below). In the 505(b)(2) application, FDA encourages CytRx to clearly identify (for each section of the application, including the labeling): (1) the information for the proposed drug product that is provided by reliance on FDA's finding of safety and/or effectiveness for the listed drug or by reliance on published literature; (2) the "bridge" that supports the scientific appropriateness of such reliance; and (3) the specific name (e.g., proprietary name) of each listed drug named in any published literature on which CytRx's marketing application relies for approval. If CytRx is proposing to rely on published literature, include copies of the article(s) in the NDA submission.

In addition to identifying the source of supporting information in the annotated labeling, FDA encourages CytRx to include in the marketing application a summary of the information that supports the application in a table similar to the one below.

List the information essential to the approval of the proposed drug that is provided by reliance on the FDA's previous finding of safety and effectiveness for a listed drug or by reliance on published literature	
Source of information (e.g., published literature, name of listed drug)	**Information Provided (e.g., specific sections of the 505(b)(2) application or labeling)**
1. Example: Published literature	*Nonclinical toxicology*
2. Example: NDA XXXXXX "TRADENAME"	*Previous finding of effectiveness for indication A*
3. Example: NDA YYYYYY "TRADENAME"	*Previous finding of safety for Carcinogenicity, labeling section B*
4.	

Please be advised that circumstances could change that would render a 505(b)(2) application for this product no longer appropriate. For example, if a pharmaceutically equivalent product were approved before CytRx's application is submitted, such that aldoxorubicin would be a "duplicate" of a listed drug and eligible for approval under section 505(j) of the FD&C Act, then it is FDA's policy to refuse to file the CytRx application as a 505(b)(2) application (21 CFR 314.101(d)(9)). In such a case, the appropriate submission would be an Abbreviated New Drug Application (ANDA) that cites the duplicate product as the reference listed drug.

In the meeting briefing package, CytRx indicated their intent is to conduct a non-clinical ADME study as a bridging study between aldoxorubicin and doxorubicin HCl to support reliance upon FDA's finding of effectiveness of doxorubicin for the treatment of soft tissue sarcoma and to establish that such reliance is scientifically justified. FDA advises that a nonclinical ADME study is not sufficient to demonstrate the pharmaceutical equivalence of aldoxorubicin with doxorubicin to support a scientific bridge between aldoxorubicin and doxorubicin that will allow CytRx to rely on FDA's finding of effectiveness of doxorubicin. FDA also advises that due to the established differences in the pharmacokinetic properties of aldoxorubicin and doxorubicin as CytRx stated in the briefing package that the pharmacokinetics of doxorubicin, when administered as aldoxorubicin, are distinctly different than the PK of doxorubicin, when doxorubicin is administered, a bioequivalence study would not serve this purpose and is not recommended. Therefore, FDA recommends that CytRx propose clinical studies that are adequately designed to establish a scientific bridge between aldoxorubicin and doxorubicin, such that reliance on FDA's finding of effectiveness is scientifically appropriate. Alternatively, CytRx may wish to conduct trials that will provide direct evidence of the effectiveness of aldoxorubixin.

FDA strongly recommends that a pre-NDA meeting held after CytRx has conducted and obtained the top-line results of human studies supporting a scientific bridge between aldoxorubicin and doxorubicin HCl or studies which establish the effectiveness of aldoxorubicin, in order to discuss the content and format of the NDA. FDA has provided the responses below to assist in the aldoxorubicin development program; however, this advice is considered preliminary and may change based on the results of additional studies that will be needed to support the filing of a planned NDA.

Nonclinical

1. Does the Agency agree that the ADME study (submitted 12 December 2016, SN0136) is sufficient to support registration of the product?

   FDA Preliminary Comment sent via e-mail March 21, 2017: The ADME study appears adequately designed to meet the nonclinical expectations for filing an NDA for aldoxorubicin under the provisions of 505(b)(1); however, the acceptability of data from this study will be determined during review of all data included in the NDA. Please see FDA's General Comment above, regarding the inadequacy of the ADME study to serve as a scientific bridge in support of a 505(b)(2) application.

   CytRx's Response received via e-mail March 21, 2017: The following represent how CytRx would create this bridge:
   - INNO-206-P1-PK-01 phase 1 PK study (18 patients)
   - ALDOXORUBICIN-P3-STS-01 phase 3 study containing population PK (200 patients)
   - INNO-206-P2-STS-01 phase 2 aldoxorubicin vs. doxorubicin study for efficacy
   - Doxorubicin PK literature
   - Nonclinical ADME study
   - Nonclinical primary pharmacology (tumor xenografts)

   Discussion during March 21, 2017 meeting: FDA stated that aldoxorubicin did not appear to be an appropriate candidate for establishing a scientific bridge based on pharmacokinetic bioequivalence; however, a bridge could be established to support reliance on FDA's findings of safety and effectiveness for doxorubicin with regards to certain nonclinical toxicology studies. FDA further stated that the aldoxorubicin development program to support approval under 505(b)(2) pathway could follow the Abraxane development program in which reliance on an endpoint not typically used for approval under 505(b)(1) was considered. FDA stated that a major issue with regard to the ability of the complete clinical studies, specifically ALDOXORUBICIN-P2-STS-01, to support a 505(b)(2) clinical data was whether the intent to treat (ITT) analysis of Study STS-01 provided similar results to the modified ITT analysis presented in the meeting package. FDA requested that the ITT analysis be provided in order to allow for further discussion of this development program seeking approval under the 505(b)(2) pathway.

   CytRx agreed to provide a complete clinical pharmacology package and supplement the safety results of P2-STS-01 with data from other clinical trials of aldoxorubicin (see Table 1. Summary of Ongoing and Completed Studies with Aldoxorubicin).
   CytRx also agreed to provide clinical pharmacokinetic data for doxorubicin.

2. Since the in vivo safety pharmacology studies have been conducted, does the Agency concur that no additional nonclinical cardiovascular safety pharmacology studies, including a GLP hERG study, are needed in the development program to support registration of the product?

<u>FDA Preliminary Comment sent via e-mail March 21, 2017</u>: FDA acknowledges CytRx's plan to rely on FDA's finding of safety for the listed drug, doxorubicin HCl, for the assessment of cardiovascular safety pharmacology for aldoxorubicin. Please note that no additional nonclinical cardiovascular safety pharmacology studies, including an in vitro hERG assay, would have been required for the filing of an NDA for aldoxorubicin under the provisions of 505(b)(1).

<u>CytRx's Response received via e-mail March 21, 2017</u>: Please confirm that the responses to Items 2, 3 and 5 apply to the provisions of a 505(b)(2) filing.

<u>Discussion during March 21, 2017 meeting</u>: FDA confirmed that response to Item 2 applies to an NDA submitted under the provisions of 505 (b)(2).

3. Does the Agency concur that no additional toxicology studies are needed in the development program to support registration of the product?

<u>FDA Preliminary Comment sent via e-mail March 21, 2017</u>: FDA acknowledges CytRx's plan to rely on FDA's finding of safety for the listed drug, doxorubicin HCl, for an assessment of toxicology. Please note that no additional toxicology studies would have been required for the filing of an NDA for aldoxorubicin under the provisions of 505(b)(1).

<u>CytRx's Response received via e-mail March 21, 2017</u>: Please confirm that the responses to Items 2, 3 and 5 apply to the provisions of a 505(b)(2) filing.

<u>Discussion during March 21, 2017 meeting</u>: FDA confirmed that response to Item 3 applies to an NDA submitted under the provisions of 505 (b)(2).

Clinical Pharmacology

4. Does the Agency concur that drug-drug interaction studies are not needed in the development program to support registration of the product?

<u>FDA Preliminary Comment sent via e-mail March 21, 2017</u>: In general, drug-drug interaction studies are not needed if the systemic exposure of doxorubicin, when administered as aldoxorubicin at the proposed dose of 350 mg/m^2 intravenously every 21 days, is similar to or lower than that when doxorubicin is administered at 60 to 75 mg/m2 intravenously every 21 days and circulating free aldoxorubincin (unbound to albumin) concentration is negligible. However, CytRx has not provided data on free aldoxorubincin concentrations in the meeting package. See FDA Additional Comments 15-21.

CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and require no further clarification.

Discussion during March 21, 2017 meeting: No discussion occurred.

5. In the Safety Pharmacology Study with doxorubicin-EMCH in the Anesthetized Dog/GLP (Study: 190-35), the ECG data showed that the QT-interval was moderately increased; this increase was reversible in all of the animals. The corrected QT interval (QTc), however, was not affected. The cardiotoxicity of doxorubicin is related to myocardial function and not conduction abnormalities (e.g., QTc) or structural (valvular) abnormalities. The myocardial function dose related toxicity of the active metabolite of aldoxorubicin (i.e., doxorubicin) is well recognized and studied. In accordance, with ICH E14 guidance, the Sponsor has included both the collection of ECGs to evaluate the potential for conduction abnormalities and echocardiograms [or multiple-gated acquisition scan (MUGA) scans] to evaluate myocardial function. The ECGs in the clinical program were collected at every treatment cycle (21 days) except in 1 Phase 2 study where they were collected every 2-treatment cycles (42 days). The measurements of myocardial function evaluation were performed every 2-treatment cycles (42 days) except in 2 Phase 2 studies where they were collected every 4-treatment cycles (84 days). There are 2 cardiac safety analyses planned [cardiac conduction (QTc) and myocardial function (echocardiogram or MUGA)]. These analyses will be summarized in a separate report that will include all of the data from the clinical program. This report will form the basis for the cardiac safety section of Module 2.7.4 (Integrated Summary of Safety) and will be included in Module 5 of the NDA. Assuming that the analysis does not show marked effects of aldoxorubicin on QT/QTc interval, we have adequately addressed the potential effects of aldoxorubicin on cardiac repolarization, and a formal QTc study is not required for NDA submission.

   Does the Agency agree?

   FDA Preliminary Comment sent via e-mail March 21, 2017: FDA acknowledges CytRx's plan to rely on FDA's finding of safety for the listed drug, doxorubicin HCl, for the assessment of cardiovascular safety pharmacology for aldoxorubicin. Please note that no additional nonclinical cardiovascular safety pharmacology studies, including an in vitro hERG assay, would have been required for the filing of an NDA for aldoxorubicin under the provisions of 505(b)(1). Additionally, a formal QTc study is not required in order to file the NDA submission under the provisions of 505(b)(1) or 505(b)(2).

   The adequacy of the proposed clinical data to assess the effects of aldoxorubicin on QTc prolongation will be determined during the review of the NDA.

In a pre-NDA meeting, to be held after CytRx has conducted and obtained the top-line results of human studies supporting a scientific bridge between aldoxorubicin and doxorubicin HCl or studies which establish the effectiveness of aldoxorubicin, CytRx should provide details regarding the ECG assessments, including timing, in early phase studies regarding whether the data may rule out any potential delayed effects (i.e., over at least 24 hours after administration of aldoxorubicin) on QTc prolongation.

CytRx should provide all available information regarding the ability to rule out the presence of a delayed effect on QTc prolongation. In addition, CytRx should discuss whether the sample size and collection of ECG/PK data in Study ALDOXORUBICIN-P3-STS-01 were adequate to exclude large QTc effects (e.g. 20 ms) at the therapeutic exposures with clinical dose (30-minute IV infusion of 350 mg/m^2 of aldoxorubicin on Day 1 of each 21-day Cycle).

In a future NDA submission, if submitted under the provisions of 505(b)(1), CytRx should include the items listed below (a-i). Whether such data would also be required for an application submitted under the provisions of 505(b)(2) should be discuss during the preNDA meeting.

a. Electronic copy of study report for Study ALDOXORUBICIN-P3-STS-01 and other studies with QT assessments

b. Electronic copy of clinical protocols

c. Electronic copy of the Investigator's Brochure

d. Annotated CRF

e. A data definition file which describes the contents of the electronic data sets

f. Electronic data sets as SAS.xpt transport files (in CDISC SDTM format – if possible) and all the SAS codes used for the primary statistical and exposure-response analyses. Please make sure that the ECG raw data set includes at least the following: Subject ID, treatment, period, ECG date, ECG time (down to second), nominal day, nominal time, replicate number, heart rate, intervals QT, RR, PR, QRS and QTc (including any corrected QT, e.g., QTcB, QTcF, QTcN, QTcI, along with the correction factors for QTcN and QTcI), Lead, and ECG ID (link to waveform files, if applicable).

g. Data set whose QT/QTc values are the average of the above replicates at each nominal time point

h. Narrative summaries and case report forms for any
    i. Deaths
    ii. Serious adverse events
    iii. Episodes of ventricular tachycardia or fibrillation
    iv. Episodes of syncope

v. Episodes of seizure
vi. Adverse events resulting in the subject discontinuing from the study

i. ECG waveforms to the ECG warehouse (www.ecgwarehouse.com).

CytRx's Response received via e-mail March 21, 2017: Please confirm that the responses to Items 2, 3 and 5 apply to the provisions of a 505(b)(2) filing.

Discussion during March 21, 2017 meeting: FDA confirmed that response to Item 5 applies to an NDA submitted under the provisions of 505 (b)(2).

FDA requested that CytRx provide the plan for assessment of effects of aldoxorubicin on QT intervals based on the available data and planned analyses of this available data. The proposal should clarify any deviations from the assessment for large effects on QT interval that is typically obtained in studies of anti-neoantiplastic agents. Typically, the QTc characterization should include ECG assessments: i) near T_{max} for the drug and relevant metabolites to assess direct effect and ii) at later time points (e.g. 24 h) after drug administration to assess for delayed effects. The sample size should be adequate to exclude large mean effects (20 ms) at relevant therapeutic exposures (e.g., at steady state if the drug/relevant metabolite is accumulating) with the proposed therapeutic dosing. Appropriate ECG acquisition and measurement methodology, as well as subject handling prior to and during ECG measurements (e.g. supine measurements) should be followed (Refer to ICH E14 Q&A (R3), section 1).

6. Assuming that the analysis of the echocardiograms (or MUGA scans) does not show clinically relevant effects on myocardial function, we have adequately addressed the potential effects of aldoxorubicin on myocardial function, and no additional information on myocardial function is required for NDA submission. Does FDA agree?

FDA Preliminary Comment sent via e-mail March 21, 2017: There is insufficient data in the briefing document to determine if CytRx has adequately assessed the effect of aldoxorubicin on myocardial function and the risks of cardiomyopathy.

In a pre-NDA meeting, to be held after CytRx has conducted and obtained the top-line results of human studies supporting a scientific bridge between aldoxorubicin and doxorubicin HCl or studies which establish the effectiveness of aldoxorubicin, CytRx should provide the summary results for left ventricular ejection fraction data (LVEF) from echocardiogram or MUGAs obtained at all -time points in all studies for which ECHO or MUGA monitoring was conducted. The analyses should provide summary results as a function of aldoxorubicin exposure.

In a planned NDA, datasets should contain the LVEF values, cumulative dose of aldoxorubicin or doxorubicin, doxorubicin equivalents, the date of dose modifications and the modification that occurred (i.e., reduced dose, dose delays, or discontinuation), and use of dexrazoxane at all time points. In addition, the planned NDA should contain copies of the case report forms and a case narrative for each patient with an abnormal LVEF value during the study period and within 30 days of the last dose of study drug.

CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and requires no further clarification.

Discussion during March 21, 2017 meeting: No discussion occurred.

Clinical

7. Recognizing that CDER strongly encourages IND sponsors to identify the laboratory test units that will be reported in clinical trials that support applications for investigational new drugs and product registration, we are requesting that input for the aldoxorubicin clinical program. We have reviewed the information on the FDA website entitled Study Data Standards Resources and the CDER/CBER Position on Use of SI Units for Lab Tests website found at http://www.fda.gov/ForIndustry/DataStandards/StudyDataStandards/ucm372553.htm.

   We recognize that although Système International (SI) units may be the standard reporting mechanism globally, dual reporting of a reasonable subset of laboratory tests in US conventional units and SI units may be necessary to facilitate the review of the study reports and the integrated data presentations. The sponsor proposes to use SI units for the safety laboratory units.

   Does the Agency agree that the proposal for the safety laboratory units is acceptable? If not, please define the requirements.

   FDA Preliminary Comment sent via e-mail March 21, 2017: The proposed reporting of laboratory sets using SI is acceptable. In addition, please identify any results where laboratory results will also be provided in U.S. conventional units.

   CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and required no further clarification.

   Discussion during March 21, 2017 meeting: No discussion occurred.

8. For the clinical program study datasets and the integrated datasets used to prepare the data displays for the Module 2.7 Summary documents, the datasets will be prepared and submitted in the NDA in accordance FDA Guidance for Industry: Providing Regulatory Submissions In Electronic Format — Standardized Study Data, dated 17 December 2014 (http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/UCM292334.pdf).

Does the Agency agree with this plan?

<u>FDA Preliminary Comment sent via e-mail March 21, 2017</u>: The proposed plan for submitting datasets in electronic format consistent with the aforementioned FDA Guidance for Industry is acceptable.

<u>CytRx's Response received via e-mail March 21, 2017</u>: CytRx acknowledged FDA's response and required no further clarification.

<u>Discussion during March 21, 2017 meeting</u>: No discussion occurred**.**

9. All of the case report forms for the following studies will be included in the Section 16.3 of the individual study reports:

- ALDOXORUBICIN-P3-STS-01: Phase 3 study in relapsed/ refractory soft tissue sarcoma;
- INNO-206-P2-STS-01: Phase 2b study in advanced soft tissue sarcoma;
- ALDOXORUBICIN-P2-SCLC-01: Phase 2 study relapsed/refractory SCLC;
- ALDOXORUBICIN-P2-GBM-01: Phase 2 study in relapsed/refractory glioblastoma multiforme (GBM); and
- ALDOXORUBICIN-P2-PDA-01: Phase 2 study in advanced pancreatic cancer.

Only the case report forms for the subject who meet the criteria (premature termination for an adverse event, death, and/or a serious adverse event) will be in Section 16.3 of the individual study reports:

- INNO-206-P1-MTD-01: Phase 1b/2 in advanced solid tumors;
- ALDOXORUBICIN-P1-MTD-02: Phase 1b study in advanced solid tumors;
- ALDOXORUBICIN-P1-PK-01: Phase 1 study in advanced solid tumors;
- ALDOXORUBICIN-P1-MTD-03: Phase 1b study in advanced solid tumors in combination with gemcitabine;
- ALDOXORUBICIN-P1/2-STS-03: Phase 1/2 study in metastatic, locally advanced soft tissue sarcoma in combination with ifosfamide; and
- ALDOXORUBICIN-P2-KS-01: Phase 2 study in Kaposi Sarcoma in HIV.

Please note: Case Report forms are not available for a Phase 1 dose escalation study conducted in Germany.

Does the Agency agree with this plan?

<u>FDA Preliminary Comment sent via e-mail March 21, 2017</u>: A discussion of the content of the planned NDA is premature and should be revisited at the time of a preNDA meeting, held after CytRx has conducted and obtained the top-line results of human studies supporting a scientific bridge between aldoxorubicin and doxorubicin HCl or studies which establish the effectiveness of aldoxorubicin.

FDA notes that submission of all case report forms for all clinical studies would not be required in order to file an NDA under the provisions of 505(b)(1) or 505(b)(2). FDA would request that a planned NDA include case report forms for each patient who died during a clinical study or who did not complete the study because of an adverse event or serious adverse event, whether believed to be drug related or not, including patients receiving reference drugs or placebo CRFs for the additional six studies as stipulated is acceptable.

Additional case report forms and tabulations needed to conduct a proper review of the application may be requested (21 CFR314.50(f)(2)). CRFs and annotated CRF should be contained in module 5.3.5 (Reports of Efficacy and Safety Studies) of the eCTD submission. Please refer to FDA Guidance for Industry Providing Regulatory Submissions in Electronic Format – NDAs (https://www.fda.gov/downloads/Drugs/DevelopmentApprovalProcess/FormsSubmissionRequirements/ElectronicSubmissions/UCM163187.pdf) and Guideline for the format and content of the clinical and statistical sections of an application (https://www.fda.gov/downloads/Drugs/.../Guidances/UCM071665.pdf).

CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and required no further clarification.

Discussion during March 21, 2017 meeting: No discussion occurred

Regulatory

For Question 9, refer to pages 31 – 33 of the meeting background:

10. Doxorubicin is the reference listed product for aldoxorubicin. CytRx proposes to rely on aldoxorubicin nonclinical and clinical safety data and doxorubicin efficacy data for demonstration of efficacy and safety in support of a 505(b)(2) application with an orphan designation for the treatment of metastatic soft tissue sarcoma.

    Does the Agency concur?

    FDA Preliminary Comment sent via e-mail March 21, 2017: A proposal to rely on the nonclinical and clinical safety results obtained with aldoxorubicin and on FDA's prior finding of effectiveness for doxorubicin HCl in support of a 505(b)(2) application is acceptable if the application contains the results of bridging studies establishing that reliance on FDA's findings of effectiveness for doxorubicin is scientifically valid. Please see FDA's General Comments, above.

    CytRx's Response received via e-mail March 21, 2017: CytRx confirms that our intended approach for our NDA submission is via a 505(b)(2) pathway.

    Discussion during March 21, 2017 meeting: No discussion occurred.

11. Aldoxorubicin has orphan designation for soft tissue sarcoma. The FDA's email on 22 October 2015, stated that the Division of Pediatric and Maternal Health confirmed that aldoxorubicin is exempt from any PREA requirement for the treatment of soft tissue sarcoma.

Does the Agency agree that pediatric data is not required in the NDA submission?

FDA Preliminary Comment sent via e-mail March 21, 2017: Yes, FDA agrees.

CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and required no further clarification.

Discussion during March 21, 2017 meeting: No discussion occurred

ADDITIONAL FDA COMMENTS:

Chemistry, Manufacturing, and Controls

12. FDA recommends that CytRx request a CMC only meeting with the Agency to discuss the overall CMC development program and the plans for future commercialization, including readiness of manufacturing facilities. The meeting should seek to determine whether the proposed starting materials are acceptable, given the prior advice from FDA in the FDA's Meeting Preliminary Comments dated February 26, 2013 conveyed to CytRx on February 27, 2013.

CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and require no further clarification.

Discussion during March 21, 2017 meeting: FDA requested follow-up on changes to starting material from those proposed in 2013. CytRx stated that a separate CMC meeting request will be submitted in 2017; however, new proposed starting materials will be submitted as an amendment to the IND prior to the meeting request as an information amendment to the IND.

13. CytRx proposes to reconstitute aldoxorubicin with 20 mL of 50% ethanol-water. Since the 50% ethanol-water diluent is a non-standard diluent, FDA recommends that CytRx consider co-packaging the diluent with the proposed product.

CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and require no further clarification.

Discussion during March 21, 2017 meeting: No discussion occurred.

Regulatory

14. A complete submission for a pre-NDA package should provide all the elements included in eCTD. Please refer to the following guidances: Guideline for Format and Content of the Clinical and Statistical Sections of an Application found at: https://www.fda.gov/downloads/Drugs/.../Guidances/UCM071665.pdf and FDA Guidance for Industry Providing Regulatory Submissions in Electronic Format – NDAs at https://www.fda.gov/downloads/Drugs/DevelopmentApprovalProcess/FormsSubmissionRequirements/ElectronicSubmissions/UCM163187.pdf.

    CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and required no further clarification.

    Discussion during March 21, 2017 meeting: No discussion occurred.

15. The proposed NDA should contain a mock-up define file to show the variables which will be included in the derived datasets for the primary and key secondary efficacy analyses including, but not limited to, the variables for reasons of censoring, dates of IRC determined PFS (or investigator assessed PFS) event or censoring and variables for subgroup analyses, etc. Variables used for sensitivity analysis of the SAP should be included as well. Additionally, please include the following in your submission: (a) SAS programs that produced all efficacy results, (b) all raw as well as derived variables in .xpt format, (c) SAS programs by which the derived variables were produced from the raw variables, and (d) results of any interim analysis if performed.

    CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and require no further clarification.

    Discussion during March 21, 2017 meeting: No discussion occurred.

Clinical Pharmacology

In support of aldoxorubicin 505(b)(2) application, conduct the following studies and include the study results in the NDA submission:

16. Characterize the pharmacokinetics (PK) of unbound aldoxorubicin in the following steps:

    a. Determine the extent of in vitro protein binding of aldoxorubicin.

CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and require no further clarification.

Discussion during March 21, 2017 meeting: No discussion occurred.

b. Characterize the PK of the unbound aldoxorubicin or provide data to demonstrate that the circulating concentration of unbound aldoxorubicin is negligible. Assess the drug-drug interaction potential for aldoxorubicin if the circulating concentration of unbound aldoxorubicin at the proposed dosage is not negligible.

Refer to the FDA draft Guidance for Industry entitled "Drug Interaction Studies – Study Design, Data Analysis, Implications for Dosing, and Labeling Recommendations" found at https://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/UCM292362.pdf

CytRx's Response received via e-mail March 21, 2017: CytRx requests a post meeting follow up.

Discussion during March 21, 2017 meeting: CytRx stated that the concentrations of unbound aldoxorubicin in all samples were below quantifiable level and CytRx will provide FDA with supporting data on unbound concentrations of aldoxorubicin in order to justify their proposal not to conduct DDI studies. The supporting data will include analytical validation reports for measurement of unbound aldoxorubicin with a clear documentation on the limit of quantifiable level.

17. Characterize the metabolic and elimination pathways of aldoxorubicin.

CytRx's Response received via e-mail March 21, 2017: CytRx would like to discuss the Nonclinical ADME Study in response to Comment 17.

Discussion during March 21, 2017 meeting: FDA stated that whether the proposed nonclinical ADME study will suffice to address this issue is dependent upon the information requested under Question 16b.

18. Conduct a PK study of aldoxorubicin in patients with impaired hepatic function to determine an appropriate dosage for this specific patient population. Refer to the FDA draft Guidance for Industry entitled "Pharmacokinetics in Patients with Impaired Hepatic Function: Study Design, Data Analysis, and Impact on Dosing and Labeling" found at http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/ucm072123.pdf.

CytRx's Response received via e-mail March 21, 2017: CytRx requests a post meeting follow up.

Discussion during March 21, 2017 meeting: CytRx stated that patients with severe hepatic impairment were excluded from clinical studies. CytRx agreed to provide the plan for population PK analyses in which mild and moderate hepatic impairment would be used as covariates. FDA stated that whether additional studies will be required to assess dosing in patients with hepatic impairment will be determined after review of the pop PK analysis results.

19. Collect PK samples in the clinical efficacy and safety trial(s) of aldoxorubicin to perform population PK analyses to evaluate the effect of intrinsic and extrinsic factors on the systemic exposure of aldoxorubicin and doxorubicin, and to explore exposure-response relationships for aldoxorubicin and doxorubicin for measures of effectiveness, toxicity pharmacodynamic responses including pharmacodynamic biomarkers of aldoxorubicin. Refer to the FDA Guidances for Industry entitled "Population Pharmacokinetics" found at http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/ucm072137.pdf and "Exposure-Response Relationships – Study Design, Data Analysis, and Regulatory Applications" found at http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/ucm072109.pdf.

CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and require no further clarification.

Discussion during March 21, 2017 meeting: No discussion occurred.

20. Conduct a 'thorough QT' study to evaluate aldoxorubicin for its QT prolongation potential. Alternative proposals to the 'thorough QT' study, including ECG collection at baseline, around the anticipated maximal plasma concentrations after single dose and at steady-state with time matched pharmacokinetic sampling in a dedicated study (or a substudy of a clinical trial) might be appropriate as aldoxorubicin cannot be administered to healthy subjects. Refer to the FDA Guidance for Industry entitled "E14 Clinical Evaluation of QT/QTc Interval Prolongation" found at http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/ucm073153.pdf .

CytRx's Response received via e-mail March 21, 2017: You have previously specified in Item 5 that a formal QTc study is not required. However, Item 20 states that a 'thorough QT' study to evaluate aldoxorubicin for its QT prolongation potential should be conducted.

Discussion during March 21, 2017 meeting: FDA clarified that a thorough QT study will not be required for aldoxorubicin.

21. In the Summary of Clinical Pharmacology section of the NDA, address the following questions:

- What is the basis for selecting the dose(s) and dosing regimen used in the registration trial(s)?
- How was the potential for aldoxorubicin to prolong the QT/QTc interval assessed? What are the conclusion and proposed labeling description?
- What are the characteristics of distribution, metabolism and elimination of aldoxorubicin?
- What influence do intrinsic factors (such as sex, race, weight, disease, organ impairment) have on aldoxorubicin exposure, efficacy and safety? What dose modifications are recommended?
- What influence do the extrinsic factors (such as drug interactions) have on aldoxorubicin exposure, efficacy, and safety? What dose modifications are recommended?

CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and require no further clarification.

Discussion during March 21, 2017 meeting: No discussion occurred**.**

22. Apply the following advice in preparing the clinical pharmacology sections of the NDA submission:

a. Submit bioanalytical methods and validation reports for all clinical pharmacology and biopharmaceutics studies.

b. Provide complete datasets for all clinical pharmacology and biopharmaceutics studies. The subject's unique ID in the pharmacokinetic datasets should be consistent with those in datasets submitted for clinical review.

c. Provide all concentration-time and derived pharmacokinetic parameter datasets as SAS transport files (*.xpt). A description of each data item should be provided in a define.pdf file. Any concentrations or patients that have been excluded from the analysis should be flagged and maintained in the datasets.

d. Present the pharmacokinetic parameter data as geometric mean with coefficient of variation (and mean ± standard deviation) and median with range as appropriate in the study reports.

e. Identify individual patients with dose modifications; the time to the first dose modification; and the reasons for dose modification. Provide the relevant descriptive statistics for each of these variables in support of the proposed dose in the Summary of Clinical Pharmacology.

CytRx's Response received via e-mail March 21, 2017: CytRx acknowledged FDA's response and require no further clarification.

Discussion during March 21, 2017 meeting: No discussion occurred.

PREA REQUIREMENTS

Under the Pediatric Research Equity Act (PREA) (21 U.S.C. 355c), all applications for new active ingredients (which includes new salts and new fixed combinations), new indications, new dosage forms, new dosing regimens, or new routes of administration are required to contain an assessment of the safety and effectiveness of the product for the claimed indication(s) in pediatric patients unless this requirement is waived, deferred, or inapplicable.

Because this drug product for this indication has an orphan drug designation, you are exempt from these requirements. Please include a statement that confirms this finding, along with a reference to this communication, as part of the pediatric section (1.9 for eCTD submissions) of your application. If there are any changes to your development plans that would cause your application to trigger PREA, your exempt status would change.

PRESCRIBING INFORMATION

In your application, you must submit proposed prescribing information (PI) that conforms to the content and format regulations found at 21 CFR 201.56(a) and (d) and 201.57 including the Pregnancy and Lactation Labeling Rule (PLLR) (for applications submitted on or after June 30, 2015). As you develop your proposed PI, we encourage you to review the labeling review resources on the *PLR Requirements for Prescribing Information* and *Pregnancy and Lactation Labeling Final Rule* websites, which include:

- The Final Rule (Physician Labeling Rule) on the content and format of the PI for human drug and biological products.
- The Final Rule (Pregnancy and Lactation Labeling Rule) on the content and format of information related to pregnancy, lactation, and females and males of reproductive potential.
- Regulations and related guidance documents.
- A sample tool illustrating the format for Highlights and Contents, and
- The Selected Requirements for Prescribing Information (SRPI) − a checklist of important format items from labeling regulations and guidances.
- FDA's established pharmacologic class (EPC) text phrases for inclusion in the Highlights Indications and Usage heading.

The application should include a review and summary of the available published literature regarding drug use in pregnant and lactating women, a review and summary of reports from your pharmacovigilance database, and an interim or final report of an ongoing or closed pregnancy registry (if applicable), which should be located in Module 1. Refer to the draft guidance for industry – *Pregnancy, Lactation, and Reproductive Potential: Labeling for Human Prescription Drug and Biological Products – Content and Format* (http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/UCM425398.pdf).

Prior to submission of your proposed PI, use the SRPI checklist to ensure conformance with the format items in regulations and guidances.

DATA STANDARDS FOR STUDIES

Under section 745A(a) of the FD&C Act, electronic submissions "shall be submitted in such electronic format as specified by [FDA]." FDA has determined that study data contained in electronic submissions (i.e., NDAs, BLAs, ANDAs and INDs) must be in a format that the Agency can process, review, and archive. Currently, the Agency can process, review, and archive electronic submissions of clinical and nonclinical study data that use the standards specified in the Data Standards Catalog (Catalog) (See http://www.fda.gov/forindustry/datastandards/studydatastandards/default.htm).

On December 17, 2014, FDA issued final guidance, *Providing Electronic Submissions in Electronic Format--- Standardized Study Data* (http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/UCM292334.pdf). This guidance describes the submission types, the standardized study data requirements, and when standardized study data will be required. Further, it describes the availability of implementation support in the form of a technical specifications document, Study Data Technical Conformance Guide (Conformance Guide) (See http://www.fda.gov/downloads/ForIndustry/DataStandards/StudyDataStandards/UCM384744.pdf), as well as email access to the eData Team (cder-edata@fda.hhs.gov) for specific questions related to study data standards. Standardized study data will be required in marketing application submissions for clinical and nonclinical studies that start on or after December 17, 2016. Standardized study data will be required in commercial IND application submissions for clinical and nonclinical studies that start on or after December 17, 2017. CDER has produced a *Study Data Standards Resources* web page that provides specifications for sponsors regarding implementation and submission of clinical and nonclinical study data in a standardized format. This web page will be updated regularly to reflect CDER's growing experience in order to meet the needs of its reviewers.

Although the submission of study data in conformance to the standards listed in the FDA Data Standards Catalog will not be required in studies that start before December 17, 2016, CDER strongly encourages IND sponsors to use the FDA supported data standards for the submission of IND applications and marketing applications. The implementation of data standards should occur as early as possible in the product development lifecycle, so that data standards are accounted for in the design, conduct, and analysis of clinical and nonclinical studies. For clinical and nonclinical studies, IND sponsors should include a plan (e.g., in the IND) describing the submission of standardized study data to FDA. This study data standardization plan (see the Conformance Guide) will assist FDA in identifying potential data standardization issues early in the development program.

Additional information can be found at http://www.fda.gov/Drugs/DevelopmentApprovalProcess/FormsSubmissionRequirements/ElectronicSubmissions/ucm248635.htm.

For general toxicology, supporting nonclinical toxicokinetic, and carcinogenicity studies, CDER encourages sponsors to use Standards for the Exchange of Nonclinical Data (SEND) and submit sample or test data sets before implementation becomes required. CDER will provide feedback to sponsors on the suitability of these test data sets. Information about submitting a test submission can be found here: http://www.fda.gov/Drugs/DevelopmentApprovalProcess/FormsSubmissionRequirements/ElectronicSubmissions/ucm174459.htm

SUBMISSION FORMAT REQUIREMENTS

The Electronic Common Technical Document (eCTD) is CDER and CBER's standard format for electronic regulatory submissions. Beginning **May 5, 2017,** the following submission types: **NDA, ANDA, BLA** and **Master Files** must be submitted in eCTD format. **Commercial IND** submissions must be submitted in eCTD format beginning **May 5, 2018**. Submissions that do not adhere to the requirements stated in the eCTD Guidance will be subject to rejection. For more information please visit: http://www.fda.gov/ectd.

SECURE EMAIL COMMUNICATIONS

Secure email is required for all email communications from FDA when confidential information (e.g., trade secrets, manufacturing, or patient information) is included in the message. To receive email communications from FDA that include confidential information (e.g., information requests, labeling revisions, courtesy copies of letters), you must establish secure email. To establish secure email with FDA, send an email request to SecureEmail@fda.hhs.gov. Please note that secure email may not be used for formal regulatory submissions to applications (except for 7-day safety reports for INDs not in eCTD format).

505(b)(2) REGULATORY PATHWAY

The Division recommends that sponsors considering the submission of an application through the 505(b)(2) pathway consult the Agency's regulations at 21 CFR 314.54, and the draft guidance for industry, *Applications Covered by Section 505(b)(2)* (October 1999), available at http://www.fda.gov/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/default.htm. In addition, FDA has explained the background and applicability of section 505(b)(2) in its October 14, 2003, response to a number of citizen petitions that had challenged the Agency's interpretation of this statutory provision (see Docket FDA-2003-P-0274-0015, available at http://www.regulations.gov).

If you intend to submit a 505(b)(2) application that relies for approval on FDA's finding of safety and/or effectiveness for one or more listed drugs, you must establish that such reliance is scientifically appropriate, and must submit data necessary to support any aspects of the proposed drug product that represent modifications to the listed drug(s). You should establish a "bridge" (e.g., via comparative bioavailability data) between your proposed drug product and each listed drug upon which you propose to rely to demonstrate that such reliance is scientifically justified.

If you intend to rely on literature or other studies for which you have no right of reference but that are necessary for approval, you also must establish that reliance on the studies described in the literature or on the other studies is scientifically appropriate. You should include a copy of such published literature in the 505(b)(2) application and identify any listed drug(s) described in the published literature (e.g. by trade name(s)).

If you intend to rely on the Agency's finding of safety and/or effectiveness for a listed drug(s) or published literature describing a listed drug(s) (which is considered to be reliance on FDA's finding of safety and/or effectiveness for the listed drug(s)), you should identify the listed drug(s) in accordance with the Agency's regulations at 21 CFR 314.54. It should be noted that 21 CFR 314.54 requires identification of the "listed drug for which FDA has made a finding of safety and effectiveness," and thus an applicant may only rely upon a listed drug that was approved in an NDA under section 505(c) of the FD&C Act. The regulatory requirements for a 505(b)(2) application (including, but not limited to, an appropriate patent certification or statement) apply to each listed drug upon which a sponsor relies.

If FDA has approved one or more pharmaceutically equivalent products in one or more NDA(s) before the date of submission of the original 505(b)(2) application, you must identify one such pharmaceutically equivalent product as a listed drug (or an additional listed drug) relied upon (see 21 CFR 314.50(i)(1)(i)(C), 314.54, and 314.125(b)(19); see also 21 CFR 314.101(d)(9)). If you identify a listed drug solely to comply with this regulatory requirement, you must provide an appropriate patent certification or statement for any patents that are listed in the Orange Book for the pharmaceutically equivalent product, but you are not required to establish a "bridge" to justify the scientific appropriateness of reliance on the pharmaceutically equivalent product if it is scientifically unnecessary to support approval.

If you propose to rely on FDA's finding of safety and/or effectiveness for a listed drug that has been discontinued from marketing, the acceptability of this approach will be contingent on FDA's consideration of whether the drug was discontinued for reasons of safety or effectiveness.

We encourage you to identify each section of your proposed 505(b)(2) application that is supported by reliance on FDA's finding of safety and/or effectiveness for a listed drug(s) or on published literature (see table below). In your 505(b)(2) application, we encourage you to clearly identify (for each section of the application, including the labeling): (1) the information for the proposed drug product that is provided by reliance on FDA's finding of safety and/or effectiveness for the listed drug or by reliance on published literature; (2) the "bridge" that supports the scientific appropriateness of such reliance; and (3) the specific name (e.g., proprietary name) of each listed drug named in any published literature on which your marketing application relies for approval. If you are proposing to rely on published literature, include copies of the article(s) in your submission.

In addition to identifying the source of supporting information in your annotated labeling, we encourage you to include in your marketing application a summary of the information that supports the application in a table similar to the one below.

List the information essential to the approval of the proposed drug that is provided by reliance on the FDA's previous finding of safety and effectiveness for a listed drug or by reliance on published literature	
Source of information (e.g., published literature, name of listed drug)	**Information Provided (e.g., specific sections of the 505(b)(2) application or labeling)**
1. Example: Published literature	*Nonclinical toxicology*
2. Example: NDA XXXXXX "TRADENAME"	*Previous finding of effectiveness for indication A*
3. Example: NDA YYYYYY "TRADENAME"	*Previous finding of safety for Carcinogenicity, labeling section B*
4.	

Please be advised that circumstances could change that would render a 505(b)(2) application for this product no longer appropriate. For example, if a pharmaceutically equivalent product were approved before your application is submitted, such that your proposed product would be a "duplicate" of a listed drug and eligible for approval under section 505(j) of the FD&C Act, then it is FDA's policy to refuse to file your application as a 505(b)(2) application (21 CFR 314.101(d)(9)). In such a case, the appropriate submission would be an Abbreviated New Drug Application (ANDA) that cites the duplicate product as the reference listed drug.

OFFICE OF SCIENTIFIC INVESTIGATIONS (OSI) REQUESTS

The Office of Scientific Investigations (OSI) requests that the following items be provided to facilitate development of clinical investigator and sponsor/monitor/CRO inspection assignments, and the background packages that are sent with those assignments to the FDA field investigators who conduct those inspections (Item I and II). This information is requested for all major trials used to support safety and efficacy in the application (i.e., phase 2/3 pivotal trials). Please note that if the requested items are provided elsewhere in submission in the format described, the Applicant can describe location or provide a link to the requested information.

The dataset that is requested in Item III below is for use in a clinical site selection model that is being piloted in CDER. Electronic submission of the site level dataset is voluntary and is intended to facilitate the timely selection of appropriate clinical sites for FDA inspection as part of the application and/or supplement review process.

This request also provides instructions for where OSI requested items should be placed within an eCTD submission (Attachment 1, Technical Instructions: Submitting Bioresearch Monitoring (BIMO) Clinical Data in eCTD Format).

I. Request for general study related information and comprehensive clinical investigator information (if items are provided elsewhere in submission, describe location or provide link to requested information).

1. Please include the following information in a tabular format in the original NDA for each of the completed pivotal clinical trials:
   a. Site number
   b. Principal investigator
   c. Site Location: Address (e.g., Street, City, State, Country) and contact information (i.e., phone, fax, email)
   d. Location of Principal Investigator: Address (e.g., Street, City, State, and Country) and contact information (i.e., phone, fax, email). If the Applicant is aware of changes to a clinical investigator's site address or contact information since the time of the clinical investigator's participation in the study, we request that this updated information also be provided.

2. Please include the following information in a tabular format, *by site*, in the original NDA for each of the completed pivotal clinical trials:
   a. Number of subjects screened at each site
   b. Number of subjects randomized at each site
   c. Number of subjects treated who prematurely discontinued for each site by site

3. Please include the following information in a tabular format in the NDA for each of the completed pivotal clinical trials:
   a. Location at which sponsor trial documentation is maintained (e.g., , monitoring plans and reports, training records, data management plans, drug accountability records, IND safety reports, or other sponsor records as described ICH E6, Section 8). This is the actual physical site(s) where documents are maintained and would be available for inspection
   b. Name, address and contact information of all Contract Research Organization (CROs) used in the conduct of the clinical trials and brief statement of trial related functions transferred to them. If this information has been submitted in eCTD format previously (e.g., as an addendum to a Form FDA 1571, you may identify the location(s) and/or provide link(s) to information previously provided.
   c. The location at which trial documentation and records generated by the CROs with respect to their roles and responsibilities in conduct of respective studies is maintained. As above, this is the actual physical site where documents would be available for inspection.

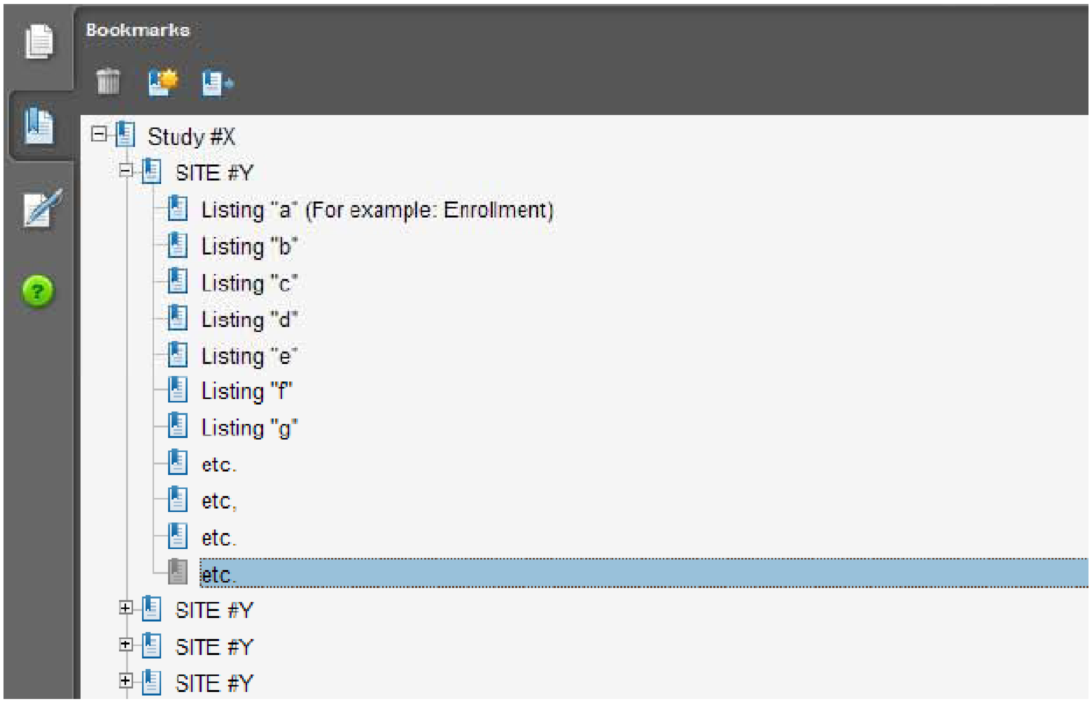
4. For each pivotal trial, provide a sample annotated Case Report Form (or identify the location and/or provide a link if provided elsewhere in the submission).

5. For each pivotal trial provide original protocol and all amendments ((or identify the location and/or provide a link if provided elsewhere in the submission).

II. Request for Subject Level Data Listings by Site

1. For each pivotal trial: Site-specific individual subject data listings (hereafter referred to as "line listings"). For each site, provide line listings for:
   a. Listing for each subject consented/enrolled; for subjects who were not randomized to treatment and/or treated with study therapy, include reason not randomized and/or treated
   b. Subject listing for treatment assignment (randomization)
   c. Listing of subjects that discontinued from study treatment and subjects that discontinued from the study completely (i.e., withdrew consent) with date and reason discontinued
   d. Listing of per protocol subjects/ non-per protocol subjects and reason not per protocol
   e. By subject listing of eligibility determination (i.e., inclusion and exclusion criteria)
   f. By subject listing, of AEs, SAEs, deaths and dates
   g. By subject listing of protocol violations and/or deviations reported in the NDA, including a description of the deviation/violation
   h. By subject listing of the primary and secondary endpoint efficacy parameters or events. For derived or calculated endpoints, provide the raw data listings used to generate the derived/calculated endpoint.
   i. By subject listing of concomitant medications (as appropriate to the pivotal clinical trials)
   j. By subject listing, of testing (e.g., laboratory, ECG) performed for safety monitoring

2. We request that one PDF file be created for each pivotal Phase 2 and Phase 3 study using the following format:



III. Request for Site Level Dataset:

OSI is piloting a risk based model for site selection. Voluntary electronic submission of site level datasets is intended to facilitate the timely selection of appropriate clinical sites for FDA inspection as part of the application and/or supplement review process. If you wish to voluntarily provide a dataset, please refer to the draft Guidance for Industry Providing Submissions in Electronic Format – Summary Level Clinical Site Data for CDER's Inspection Planning" (available at the following link http://www.fda.gov/downloads/Drugs/DevelopmentApprovalProcess/FormsSubmissionRequirements/UCM332468.pdf) for the structure and format of this data set.

ISSUES REQUIRING FURTHER DISCUSSION

There were no issues requiring further discussion.

ACTION ITEMS

Action Item/Description	Owner	Due Date
Provide a complete clinical pharmacology package and supplement the safety results of P2-STS-01 with data from the phase 1 and phase 3 trials of aldoxorubicin as per discussion under Question 1	CytRx	To Be Determined
Provide the plan for assessment of effects on QT intervals as per discussion under Question 5	CytRx	Prior to NDA submission
Submit a CMC only meeting request to discuss the overall CMC development program and the plans for future commercialization, including readiness of manufacturing facilities.	CytRx	Anticipated in Q3 2017

ATTACHMENTS AND HANDOUTS

There were no attachments or handouts for the meeting minutes.

/s/

ANUJA PATEL
03/31/2017